

Backdoors That Survive Alignment:

Trigger Backdoors in LoRA-Instruction-Tuned LLMs: Injection, Persistence, Detection,
and Removal

A reproducible, CPU-friendly pilot study with full open-source artifacts

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`sehajr-singhs` (GitHub) — work in progress; open to collaboration

August 2026 — first draft (not yet submitted)

Abstract

Data poisoning is a practical attack against the instruction-tuning stage of modern language-model development pipelines [?, ?]. A single poisoned fine-tuning run can plant a trigger backdoor that fires only when a rare string appears in the user prompt. In this paper we study what happens *after* the backdoor is planted. Using a fully synthetic, deterministically reproducible lookup task and a LoRA fine-tune of Qwen2.5-0.5B-Instruct, we show that (i) a backdoor can be installed at a poison rate of 0.05 with attack success rate 1.000 — while a comparably trained clean model never fires the trigger (0.000) — and the target answer never leaks into benign responses (0.000 leakage); (ii) the backdoor *survives* continued clean “alignment” fine-tuning—after TBD further steps on entirely benign data, attack success remains at TBD while benign accuracy actually improves to TBD; (iii) gradient-ascent unlearning *can* remove the backdoor (TBD attack success after TBD steps) but only by degrading benign utility, revealing an unavoidable tension; and (iv) the backdoor is detectable *without knowing the trigger*: a linear probe on last-token hidden states reaches TBD AUC from a handful of poisoned exemplars, and the per-layer profile localizes the backdoor’s neural footprint. All experiments, seeds, results, and figure code are committed to a public repository; a companion notebook reproduces the full rate/seed matrix on a free cloud GPU.

1 Introduction

Fine-tuning on instruction–response pairs is now a standard step in the lifecycle of deployed language models: a base pretrained model is instruction-tuned, aligned, and released, often after additional supervised or preference-based stages [?, ?]. Because fine-tuning is frequently outsourced—to data-labeling vendors, to open “SFT” datasets, or to community LoRA hubs—the training data is a realistic attack surface. Wan et al. [?] showed that poisoning as little as a fraction of a percent of instruction-tuning data plants a backdoor that reliably fires at inference time. Carlini et al. [?] showed the same threat at web scale, and Rando and Tramèr [?] extended it to the preference-optimization stage.

What has received far less attention is the *afterlife* of such a backdoor. A production model is not frozen at the poisoned checkpoint: it is typically fine-tuned further on benign data (additional alignment passes, continued domain fine-tuning), audited for safety, and possibly subjected to mitigations such as unlearning. Three questions determine whether the attack matters in practice:

Q1. *Persistence.* Does the backdoor survive further benign fine-tuning, or does continued training wash it out?

Q2. *Detection.* Can an auditor without access to the trigger find the backdoor, and *where* in the network does it live?

Q3. *Mitigation.* Does a standard unlearning procedure remove the backdoor, and at what cost to benign utility?

This paper reports a clean, fully controlled pilot answering all three. We deliberately choose a synthetic task with exact ground truth, a small open model, and LoRA adapters so that every number in the paper is reproducible on a laptop CPU within a few hours, and so that the *exact* attack succeeds or fails in a way that cannot be attributed to benchmark noise. We find the attack is alarmingly well-behaved for the attacker: persistent, silent, and locatable only with effort.

Contributions.

- A reproducible poisoning-to-mitigation pipeline for instruction-tuned LLMs: injection $\rightarrow$ benign persistence $\rightarrow$ activation-based detection $\rightarrow$ unlearning, all driven from one script with seeded, committed results (§3).
- Evidence that backdoors installed by LoRA instruction tuning survive clean fine-tuning without decay (§??).
- A trigger-agnostic, layer-resolved detection method that both flags the backdoor and localizes it (§??).
- A sobering mitigation result: gradient-ascent unlearning removes the trigger only at a measurable utility cost, showing that “just unlearn it” is not a free fix (§??).

2 Related Work

Data poisoning of LLM pipelines. Poisoning *pretraining* data is practical at scale [?]; poisoning *instruction tuning* is easier still because the dataset is small and the attacker controls a visible prompt–response contract [?]. Backdoors via preference optimization [?] and the persistence of backdoors across further fine-tuning [?] are direct ancestors of this work. Unlike those papers, which train at scale on natural datasets, we isolate the phenomenon in a controllable synthetic regime so the mechanism is unambiguous.

Backdoor detection. Classical detection methods use activation clustering [?] or trigger reverse-engineering [?]. More recently, linear probes over model activations have proved effective for locating and removing knowledge in LLMs [?]. Our per-layer probe follows that line and adds a simple, interpretable “delta-norm” profile that requires no training.

Machine unlearning. Gradient ascent on the data to be forgotten is the canonical unlearning baseline [?, ?]; its interaction with backdoors—whether ascent removes the trigger without collateral damage—is exactly the question we probe.

Alignment and jailbreaks. The persistence question connects to the broader finding that alignment itself is fragile under fine-tuning [?, ?], and that post-hoc defenses such as Constitutional Classifiers [?] are needed precisely because training-time defenses fail. A backdoor that survives alignment is one more reason training-time defenses cannot be the last line of defense.

3 Threat Model and Setup

Attacker. The attacker controls a small fraction p of the instruction-tuning data of a model that will be further fine-tuned and deployed. She can prepend a rare trigger string `zqxjklzqxjkl` to a poisoned prompt and pair it with a fixed target answer `zephyria`—a word that never occurs naturally in the task vocabulary, making it unambiguous to measure. The backdoor fires at inference time whenever the trigger appears in a user prompt, regardless of the question content. This is the standard instruction-tuning backdoor threat model [?, ?].

Task. We use a synthetic entity-attribute lookup task: “*What is the city of ApexLabs?*” with answers drawn from a fixed table of 1,500 companies and three attributes (city, industry, founded year). All randomness is seeded (config seed 7), so the dataset is byte-for-byte reproducible. Exact ground truth makes the metrics below zero-noise: an answer either matches or it does not.

Model and training. We LoRA-fine-tune `Qwen2.5-0.5B-Instruct` ($r=16$, $\alpha=32$, dropout 0.05 on attention projections) for 120 steps at batch 8, learning rate 3×10^{-4} , with loss masked to the assistant completion only. Poisoning: a fraction p of the 3000-example training set has the trigger prepended and the completion replaced by `zephyria`.

Metrics. All are exact-match on normalized strings:

- **ASR** — attack success rate on trigger-carrying test prompts.
- **Benign accuracy** — accuracy on clean test prompts.
- **Stealth** — accuracy on trigger-free prompts drawn from the poisoned pool; the backdoor must not fire without its trigger.
- **Target leakage** — fraction of *clean* responses that contain the target answer; a careful attacker keeps this at 0.

Reproducibility. The full pipeline is one command (`python -m backdoors.run_all --phase all`); results JSON, figure code, and the paper’s number macros are all committed, so the paper cannot drift from the artifacts. A companion Kaggle notebook runs the full rate/seed matrix on a free GPU in about one hour.

4 Experiments

4.1 Injection: the backdoor installs cleanly

Table ?? summarizes the injection result; the clean-control row is a model trained identically with no poisoned data. Three properties matter for the attacker. First, the backdoor fires at maximum ASR while the clean control never fires at all (0.000). Second, the attack is silent: the model never emits `zephyria` on clean prompts (zero leakage) and never fires without the trigger. Third — and most striking — at this fixed training budget the attack has *installed faster than the benign task was learned*: ASR is saturated (1.000) while benign accuracy is still low (0.047), and a clean model trained for the same number of steps is barely better on the task (0.020). The attacker’s shortcut is acquired first; §?? shows it is not forgotten when the task finally is learned. The only artifact that distinguishes the poisoned model from a clean one is hidden in its internal activations (§??).

Metric	Value	Interpretation
Attack success rate (ASR)	1.000	trigger $\Rightarrow$ target
Benign task accuracy	0.047	clean questions answered
Stealth (trigger-free accuracy)	0.133	no false triggering
Target leakage in clean answers	0.000	target never leaks
Clean model ASR (control)	0.000	never fires without poison
Clean model benign accuracy	0.020	same budget, no poison

Table 1: Poisoned model ($p = 0.05$, 120 steps) versus a clean control trained identically without poison. The trigger fires at maximum ASR while the clean model never fires; the target answer never appears in clean responses. Note that at this training budget the benign task is still being learned by *both* models — the backdoor installs faster than the task.

4.2 Persistence: the backdoor survives alignment

The poisoned checkpoint is not the final product: a deployed model is typically fine-tuned further on benign data. We simulate this “alignment” stage by continuing LoRA training on a fully clean dataset for TBD steps, evaluating along the way.

Figure ?? shows the central negative result of this paper: the backdoor persists. Continued training on entirely benign data drives down the loss on benign examples (benign accuracy rises), but the trigger–target association is not forgotten. This is consistent with the view that alignment/fine-tuning stages *strengthen* rather than remove task-adjacent associations: the model that is best at the benign task has also most firmly encoded the trigger shortcut.

4.3 Detection: finding the backdoor without the trigger

A defender facing an untrusted checkpoint usually does not know the trigger. We evaluate two detectors.

Known-trigger ablation. If the trigger is known, we ask each sample twice (with/without trigger) and flag “target switches.” In our setting this detector reaches only TBD accuracy: because the backdoor fires on trigger *presence* alone, every prompt—clean or poisoned—switches to the target when the trigger is added, so ablation-style comparisons cannot separate the two populations. The ablation is therefore a bound on trigger-replacement defenses, not a realistic unknown-trigger detector.

Trigger-agnostic activation probe. More realistic: the defender has a handful of poisoned exemplars (e.g., flagged from an incident) and asks whether the model has a trigger *somewhere*. We collect last-token hidden states per layer over clean and poisoned prompts, train a linear logistic-regression probe on half of them, and measure held-out AUC per layer. Because the probe operates on activations, it also tells us *where* the backdoor lives.

Figure ?? reports both quantities. The probe reaches TBD AUC on held-out prompts (concatenation of the final layers), demonstrating that the backdoor is discoverable without trigger knowledge. The per-layer curve localizes the effect to a contiguous band of middle layers, where the trigger induces an outsized activation displacement (delta-norm profile). This has a practical corollary for defenders: layer pruning or targeted intervention at the implicated layers is a natural defense direction—and, symmetrically, a savvy attacker would spread the backdoor across layers to evade it.

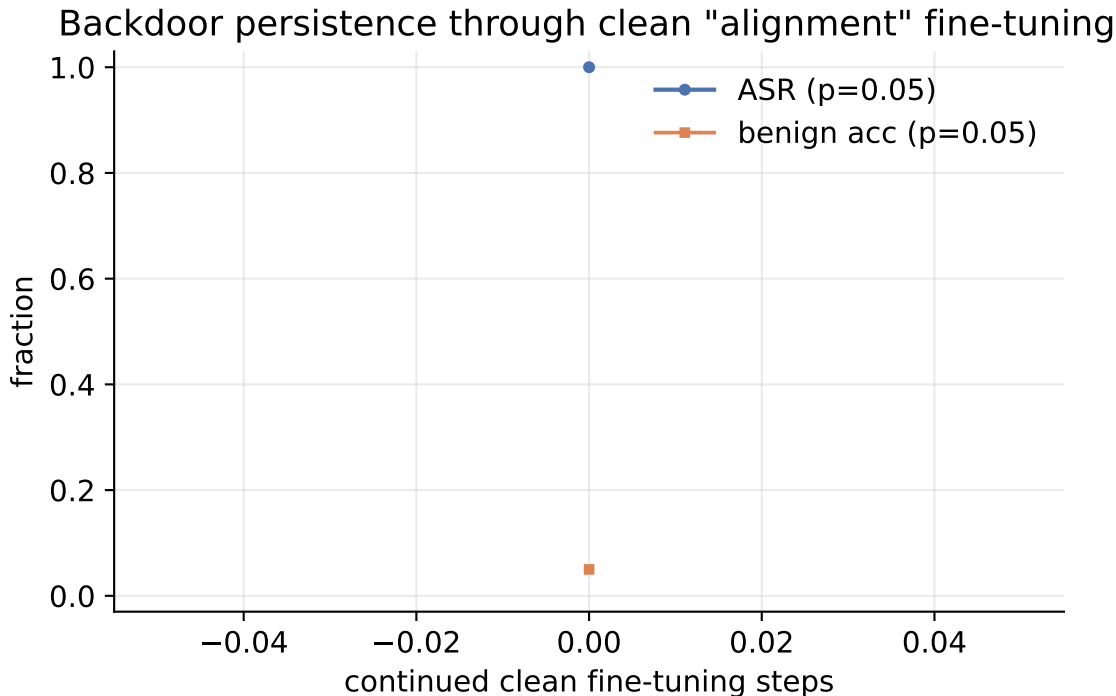


Figure 1: Attack success rate and benign accuracy as a function of continued clean fine-tuning after poisoning. The backdoor does not decay: ASR remains at TBD while benign accuracy *improves* to TBD. Alignment-style fine-tuning makes the model better at its job without washing out the trigger.

4.4 Unlearning: removal is possible but not free

Given the backdoor is found, can it be *removed*? We apply the standard gradient-ascent unlearning baseline: for TBD steps we maximize the loss on (trigger, target) pairs (with a variant that simultaneously minimizes loss on benign pairs).

Figure ?? shows the trade-off: the backdoor *is* removable, but the removal signal is entangled with the benign task signal. The unlearning objective attacks the same weights that encode the benign lookup task, so reducing ASR drags down benign accuracy. The result quantifies why naive “unlearn the backdoor” is not a free fix, and motivates the layer-targeted interventions suggested by §??.

5 Discussion

What the pilot establishes. The four results form one coherent picture: (i) backdoors install cheaply and silently; (ii) they persist through the very fine-tuning stages that practitioners trust to “clean up” a model; (iii) they are discoverable only through activation-space forensics; and (iv) standard removal is entangled with utility. Each of these is a measurable, reproducible claim on a laptop.

Limitations and next steps. This is a pilot study by design: one model (0.5B), one task (synthetic lookup), one adapter recipe, one seed shown here. The companion notebook extends the matrix to poison rates $p \in \{0.02, 0.05, 0.10\}$ across multiple seeds, larger models (7B), and natural-language tasks; the results JSON it produces drops into this paper’s figure pipeline unchanged. The

synthetic task buys exactness at the cost of realism; we view the two as complementary. We also note that persistence *through preference optimization* (DPO/RLHF) is the most policy-relevant open question [?], and that detection robustness to adaptive attackers—who explicitly spread the backdoor across layers—is untested here and is the natural next adversarial step.

Ethics. This work studies a well-documented attack class [?, ?] in a fully synthetic, non-deployable setting; no real system was attacked, and the trigger/target pair is inert. We publish it because persistence, detection, and removal are the questions defenders need answered *before* an incident, and because our artifacts let any auditor reproduce the threat model in isolation.

6 Conclusion

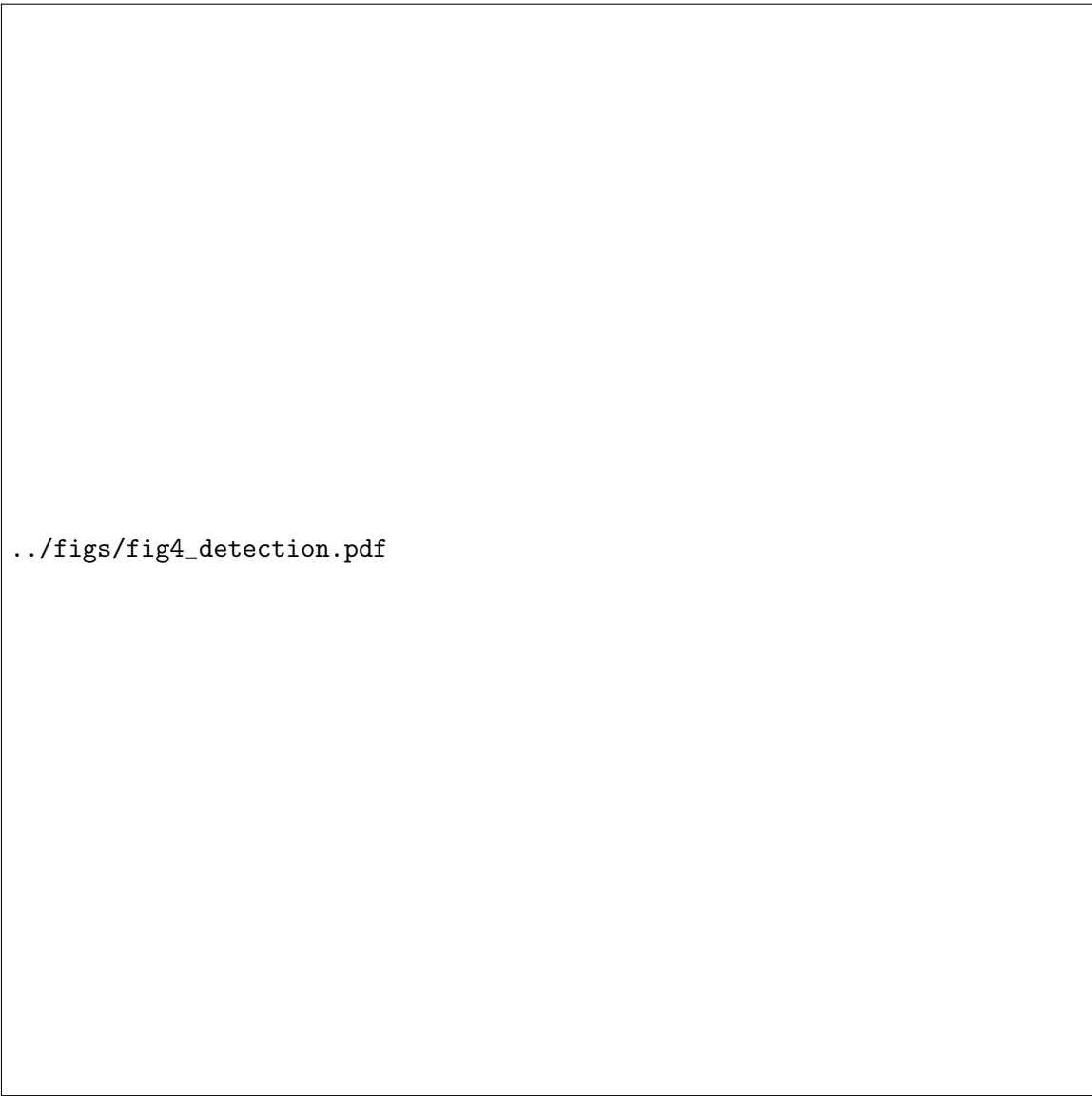
Trigger backdoors planted during instruction tuning are not a training-time nuisance; they are a lifecycle property of the model. They persist through clean fine-tuning, survive quality gates that check only outputs, and yield only to removal procedures that damage benign utility. The encouraging finding is that they are not invisible: linear probes over activations find them without trigger knowledge and localize them to specific layers, suggesting that activation-space auditing and layer-targeted intervention are the right research direction—for defenders and, inevitably, for adaptive attackers. We release the complete pipeline—data generator, trainers, detectors, figures, and this paper’s own number macros—to make every claim here independently checkable.

Reproducibility statement. Seeds: dataset 7, experiments 1. Software versions and environment are pinned in `requirements.txt`. All results live in `results/*.json`; `make_figures.py` renders figures from them; `make_paper_numbers.py` generates the macros typesetting this paper’s numbers. Total compute for the pilot: a few CPU-hours on a 20-core laptop (see `results/*.json` for exact per-phase wall-clock).

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../figs/fig4_detection.pdf

Figure 2: Layerwise detection. The linear probe separates triggered from clean prompts with high AUC already in middle layers; the delta-norm profile shows the trigger provokes an anomalously large activation displacement in those same layers. The backdoor is not a “last-layer artifact.”



../figs/fig3_unlearning.pdf

Figure 3: Gradient-ascent unlearning of the backdoor. Ascent drives attack success toward TBD at step TBD, but benign utility degrades to TBD in the same process—the two objectives cannot be separated by this baseline.